DFS

Time limit: 1 s *Memory limit: 256 MiB*

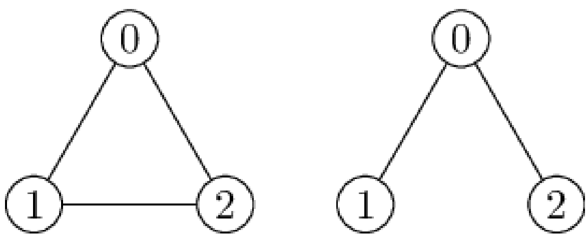
You might already be familiar with the famous DFS (depth-first search) algorithm for traversing a graph. In this problem, we will only consider connected undirected simple graphs (without loops and parallel edges) with vertices numbered 0, 1, . . ., $n - 1$, and the DFS algorithm will output depths and vertices as follows:

```
DFS(d, v):
    output d/v
    mark vertex v as visited
    W = the list of neighbors of v ordered by increasing numbers
    for each w in W:
        if vertex w has not yet been visited:
            DFS(d + 1, w)
```

Write a program that outputs the number of different graphs for which the call `DFS(0, n - 1)` produces the same printout as the one given on the input. For example, the output

```
0/2
1/0
2/1
```

is produced by calling `DFS(0, 2)` on either of the following two connected undirected simple 3-vertex graphs:



Input

The input is the output of the call `DFS(0, n - 1)` on an unknown n-vertex connected undirected simple graph. The input thus consists of n lines in the format `d/v`, with the first line being `0/n-1`.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$

Subtasks

- Subtask 1 (10 points): $n \leq 6$.
- Subtask 2 (20 points): $n \leq 500$.
- Subtask 3 (20 points): $n \leq 10^4$.
- Subtask 4 (10 points): For each $i \in \{2, \ldots, n\}$, the i-th input line is `i-1/i-2`.
- Subtask 5 (20 points): For each $i \in \{2, \ldots, n\}$, the i-th input line is `i-1/v` for some $v \in \{0, \ldots, n-2\}$.
- Subtask 6 (20 points): No additional constraints.

Output

Print the number of different graphs with the required property. Because this number can be very large, output the result modulo 1 000 000 007.

Example

Input	Output

`0/2`
`1/0`
`2/1`
	`2`